STARWEAVER®

Hands-On-Learning: Evaluating Adversary Behavior and Attribution

Learning Objectives

By the end of this activity, learners will be able to:

- Translate validated intelligence into operational security controls and defensive actions.
- Map threat intelligence findings to defensive frameworks such as detection rules or ATT&CK-based mitigation strategies.



- Develop a clear workflow for routing intelligence to appropriate operational teams.
- Evaluate how intelligence supports detection engineering, incident response, or risk reduction initiatives.
- Produce a short operational briefing demonstrating how intelligence improves measurable security posture.



Description

In this hands-on activity, learners will simulate how intelligence about adversary behavior can be used to guide operational security decisions. Participants will translate observed behaviors into actionable intelligence using the MITRE ATT&CK framework. They will then evaluate how their analysis helps mitigate the potential impact of an attack. The goal is to refine the ability to map ambiguous activity to precise adversary tactics, techniques, and procedures (TTPs) and recommend defensive actions based on this structured analysis.

Structure of the Project

Title:
Evaluating Adversary Behavior and Attribution

Tools used:

- MITRE ATT&CK Navigator (conceptual or actual, optional)
- Provided adversary behavior summary
- Simple table or whiteboard

Scenario

Your organization supports a mission-critical application that handles sensitive operational and customer data. Over the past week, security teams have observed increased scanning activity against public-facing infrastructure, coinciding with industry reporting about widespread exploitation of file transfer and application vulnerabilities. While no definitive breach has been confirmed, leadership is concerned that opportunistic activity may conceal more serious compromise.

The threat intelligence team has been asked to move beyond general awareness and provide decision-relevant insight. Leadership does not want a list of vulnerabilities or a collection of indicators. They want to understand what adversary behavior would matter most, how it would likely unfold, and whether current defensive visibility is sufficient to detect and respond.

Objective

Your objective is to evaluate observed adversary behaviors and map them to MITRE ATT&CK techniques. This will help build defensive strategies, refine detection efforts, and identify gaps in current operations.



Provided Adversary Behavior Summary

You are given the following simplified observations:

- Exploitation of a public-facing application is suspected
- Evidence suggests a server-side foothold consistent with a web shell
- Unusual backend data access occurred over a short time window
- Outbound data transfer to an external service was observed

Instructions for Learners

Task 1 – Build a TTP Map:
Action: Using ATT&CK concepts, map the observed behaviors to techniques. For each technique, note whether the behavior is observed or inferred and assign a confidence level (High / Medium / Low).

Task 2 – Identify Defensive Detection Opportunities:
Action: Identify one behavioral detection opportunity and one type of telemetry required for each phase of the TTP chain.

Task 3 – Draft a Defensive Use Case Summary:
Action: Write a short defensive use case that includes:

- What behaviors matter most
- How multiple weak signals could correlate into high confidence
- What action should be taken if detections trigger

Organize Information Following This Structure for Peer Review:

1. **Title Page:**

   Activity title, your name, date.

2. **AI Tool and Process Overview:**

   Describe the ATT&CK Navigator tool used, and the key steps taken to generate the TTP mapping and defensive use case.

3 Creative Process and Team Collaboration:

Summarize how your team or group discussed and refined the adversary behaviors, built the TTP map, and identified relevant defensive actions.

2 Reflections and Future Applications:

Explain how intelligence improved decision-making efficiency and discuss how these skills could be applied in real-world threat analysis scenarios.

Instructions for Documenting and Sharing the Project for Peer Review

Document the Project

- Use word processing software to create your project summary and reflections.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report and presentation in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.

This activity gives learners a chance to apply Module 3 principles by translating ambiguous adversary behaviors into actionable intelligence. It reinforces that intelligence is valuable when it helps structure defensive responses and informs decision-making in real-time threat environments.